



## LOB 1036 - Geometria Analítica

Lista de exercícios 3 - Parte 1  
2º semestre de 2025

1.

- a)  $(-3,0)$                       b)  $(-1,-1)$                       c)  $(2, -2\sqrt{3})$                       d)  $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

2.

- a)  $\left(\sqrt{2}, \frac{7\pi}{4}\right)$                       b)  $\left(2, \frac{5\pi}{6}\right)$                       c)  $\left(2\sqrt{2}, \frac{\pi}{4}\right)$                       d)  $(5, \pi)$

3.

- a)  $r = |a|$                                               d)  $r^2 = 4\cos(2\theta)$   
b)  $r = \frac{2}{1 - \cos\theta}$                                               e)  $r = \frac{3a \sin(2\theta)}{2(\sin^3\theta + \cos^3\theta)}$   
c)  $r = 6 \sin \theta$

4.

- a)  $(x^2 + y^2) = 4xy$                                               d)  $x = -1$   
b)  $(x^2 + y^2)^3 = x^2$                                               e)  $4x^2 - 5y^2 - 36y + 36 = 0$   
c)  $y = x \operatorname{tg}(x^2 + y^2)$

6.  $(x'_1, x'_2, x'_3) = \left(2, \frac{3 + \sqrt{3}}{2}, \frac{3\sqrt{3} - 1}{2}\right)$

7.  $R = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{2}{\sqrt{2}} & \frac{\sqrt{2}}{2} \end{pmatrix}$

8.

- a)  $x + y = 2$                                               d)  $(x - 1) = (3 - y)^2$   
b)  $y - 2x = 4$                                               e)  $x^2 - y^2 = 1$   
c)  $x + y = 3$                                               f)  $2x^2 + y = 1$

9. Não.